

## Assignment 2: Credit, Derivative, Liquidity, and Operational Risk

**Due: Friday, July 17, 2026 @23:59**

Objective: model the other major risks facing your three companies, namely credit risk (default and portfolio loss), derivative and option risk, liquidity risk and operational risk.

Required tasks:

- 1) Fetch each company's market cap (equity value  $E$ ) and total debt (debt face  $D$ ). Using  $E$ ,  $D$  and your Assignment 1 equity volatility, apply the Merton structural model to compute each firm's distance-to-default and  $PD = N(-DD)$ . Explain the model and any approximation you use and set each firm's result against its agency rating or credit spread.<sup>1</sup>
- 2) Build a small loan book exposed to the three firms (e.g. EAD proportional to each firm's debt,  $LGD = 45\%$ ), compute the expected loss, then run a Monte Carlo simulation of portfolio loss with a single-factor (Gaussian-copula) default correlation. Report the loss distribution, Credit VaR (at 99% and 99.9%) and Expected Shortfall, and show how economic capital responds as you vary the asset correlation (say 0.05, 0.15 and 0.30). Discuss how this single Gaussian-copula factor can understate the chance of many firms defaulting together in a crisis and relate your simulated loss distribution to the analytic single-factor formula that sits behind the Basel capital charge.
- 3) Derivatives: price a 3-month at-the-money option on Stock 1 (using its own volatility) by Black-Scholes and by a binomial tree and check that the two converge. Report the Greeks (delta, gamma, vega), then compute the option-position VaR by a delta-gamma approximation and by full revaluation and explain why a delta-only VaR misstates the risk and how that flips for a short position.
- 4) Compute a liquidity-adjusted VaR by adding a liquidity cost to your Assignment-1 VaR, using a bid-ask spread or an illiquidity measure built from your own price and volume data (for example average absolute return per unit of dollar volume). Compare the liquid and

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<sup>1</sup> Expect tiny Merton PDs and use that: for healthy large-cap firms the structural 1-year PD is often almost zero. That is a real, reportable finding, so discuss why it happens (low leverage, and the model's thin Normal tail). To keep the loss simulation from being degenerate, floor your PDs with a rating- or CDS-based mapping. For example, map each firm to an agency rating and its historical 1-year default rate, or use a CDS-implied PD, then run the Monte Carlo on those. Comparing the structural PD with the market or rating-based PD is itself a key insight.

liquidity-adjusted figures and say which of your names is least liquid and why that matters in a sell-off.

- 5) Simulate an operational-loss distribution with a Poisson loss frequency and a lognormal severity (justify your parameters), report a 99.9% operational-loss or capital figure, and discuss a real operational-risk event relevant to one of your firms or its sector.
- 6) Discuss how Basel treats credit, market and operational risk (expected versus unexpected loss, the role of correlation, the standardized and model-based approaches) and the limitations of each model for your firms.

Deliverable: Jupyter section: methodology, simulation code, outputs, visualizations, discussion.  $\leq 10,000$  chars.